

## Performance Testing

|                      |                      |
|----------------------|----------------------|
| <b>Date</b>          | 08 October 2026      |
| <b>Team ID</b>       | UNM1611GUFRAN        |
| <b>Project Name</b>  | AI FAQ Assistant API |
| <b>Maximum Marks</b> | 5 Marks              |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Testing Tool Used</b>   | Postman                                                                                                           |
| <b>Type of Testing</b>     | Load Testing and Performance Testing                                                                              |
| <b>Target Module / API</b> | AI FAQ Assistant REST APIs – Authentication, FAQ Management, FAQ Search, AI Answer Generator and AI FAQ Generator |
| <b>Test Environment</b>    | Local DevelopmentEnvironment                                                                                      |
| <b>Test Date</b>           | 28 th September 2026                                                                                              |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description                  | No. of Virtual Users | Duration (sec) | Expected Outcome                                                            |
|------|----------------------------------------------|----------------------|----------------|-----------------------------------------------------------------------------|
| 1    | Test User Login API with multiple requests   | 10                   | 60             | User should be authenticated successfully and JWT token should be generated |
| 2    | Test Get All FAQs API with multiple requests | 20                   | 60             | FAQ records should be retrieved successfully with acceptable response time  |

|   |                                                       |    |    |                                                                                      |
|---|-------------------------------------------------------|----|----|--------------------------------------------------------------------------------------|
| 3 | Test FAQ Search API using different keywords          | 20 | 60 | Relevant FAQ results should be returned successfully without errors                  |
| 4 | Test AI Answer Generator API with different questions | 10 | 60 | AI-generated answers should be returned successfully within acceptable response time |



### Step 3: Performance Test Results

| S.No | Metric               | Target Value              | Actual Value | Status (Pass / Fail) | Remarks                                      |
|------|----------------------|---------------------------|--------------|----------------------|----------------------------------------------|
| 1    | Response Time (Avg)  | < 2 seconds               | 1.42 seconds | Pass                 | Average response time is within the target   |
| 2    | Response Time (Max)  | < 5 seconds               | 3.86 seconds | Pass                 | Maximum response time is within the target   |
| 3    | Throughput (Req/sec) | Stable request processing | 8.5 Req/sec  | Pass                 | API handled requests successfully            |
| 4    | Error Rate           | < 1%                      | 0.4%         | Pass                 | Error rate is within the acceptable limit    |
| 5    | CPU Utilization      | < 80%                     | 62%          | Pass                 | CPU utilization remained within the limit    |
| 6    | Memory Utilization   | < 80%                     | 58%          | Pass                 | Memory utilization remained within the limit |

### Step 4: Observations & Analysis

The performance testing was conducted for the AI FAQ Assistant API using Postman. The testing covered authentication, FAQ management, FAQ search and AI-based services. The APIs successfully processed multiple requests under the tested workload. The average and maximum response times were within the specified limits. The error rate was also within the acceptable range. CPU and memory utilization remained stable during testing. The AI Answer Generator may require comparatively more response time because it communicates with the Google Gemini API. Overall, the system demonstrated stable and acceptable performance under the tested conditions.

### Step 5: Screenshots / Evidence

